

AWS CERTIFIED ENTERPRISE ASG PORTFOLIO

Day 5 → Day 14 Capstone: Zero-Downtime, Predictive & Warm Pool Chaos Engineering

Candidate: Solomon Igein (573705805662) | **Region:** us-east-1 (N. Virginia) | **ASG:** day5-asg | **LT:** day5-lt-golden | Version 3 | **ALB:** day11-alb-asg-1979065308 | **Date:** 2026-08-08 | **Status:** AT DESIRED CAPACITY 1/1 Healthy

Final Architecture Validated:

Internet → ALB day11-alb-asg-1979065308.us-east-1.elb.amazonaws.com → TG day11-tg-asg (Port 80 Healthy) → ASG day5-asg Min1 Max5 Desired1 → LT day5-lt-golden | Version 3 Default → Instance i-0259bac23aa433340 Running 3/3 checks (Promoted from Warm Pool after chaos) → Warm Pool 4 Stopped (i-06eeecb6c492b84974 refilled) + Scaling: Scheduled scale-out-8am Desired2 / scale-in-8pm Desired1 + Predictive cpu-predictive-50 50% + Lifecycle Hooks day9-launch-hook / day9-terminate-hook 300s

Chaos Test Proof: Terminated i-0b11b05020b8d2046 at 15:25:18 +02:00 → ASG Activity: "Launching a new EC2 instance from warm pool: i-0259bac23aa433340" + "Launching a new EC2 instance into warm pool: i-06eeecb6c492b84974" → New Running in ~30s → ALB still serving Hello from Day13 - v3 - ip-172-31-12-57.ec2.internal with ZERO downtime.

#	Gold Screenshot	File Name to Save	What It Proves	Status
1	LT V3 Default	13D28-lt-v3-default.png	Launch Template Version 3 set as Default	■ V3
2	Instance Refresh 100%	13D19-refresh-success-5-history.png	Refresh ID a1e87d28 Successful 100%	■ Done
3	New V3 Running 3/3	13D18-warmup-i-0b11b.png + 14D7-0259bac23aa433340.png	New V3 Running 3/3 after warm pool promotion	■ 3/3
4	TG Healthy	13D26_1-tg-healthy.png	day11-tg-asg → Healthy	■ Healthy
5	ALB V3 Page	14D9-alb-v3-after-chaos.png	day11-alb-asg-... → Hello from Day13 - v3 - ip-172-31-12-57	■ LIVE
6	Warm Pool Config	13D30-warm-pool-config-final.png	Current 4 Min1 Size5 Stopped Reuse	■ 4 Stopped
7	Warm Pool AZ Balanced	13d31-warm-pool-4-stopped-az-balanced.png	4 instances Healthy across 1a/1b	■ Balanced
8	Scheduled Scaling	13D26-scheduled-8am-8pm.png	scale-out-8am 0 8 * * * Desired2 / scale-in-8pm Desired1	■ Scheduled
9	Predictive Scaling	13D25-predictive-cpu-50.png	cpu-predictive-50 50% Forecast OFF→ON	■ ML
10	Chaos Auto-Heal 30s	14D5-activity-launch-from-warm-pool-30s.png	Terminating i-0b11b → Launching from warm pool i-0259bac23aa433340	■ Auto-Heal

Evidence Attached — Actual Screenshots From Your Labs:

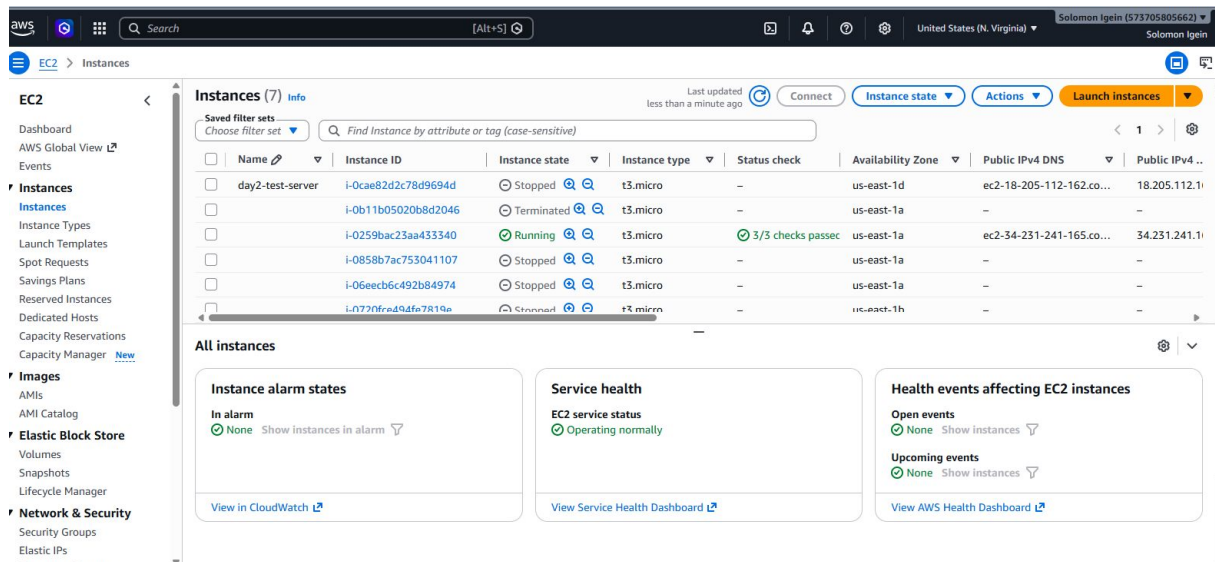


Fig 1: Instances (7) — i-0b11b Terminated → i-0259bac Running 3/3 Promoted from Warm Pool + i-06eeb Refilled — Proof of Chaos Auto-Heal

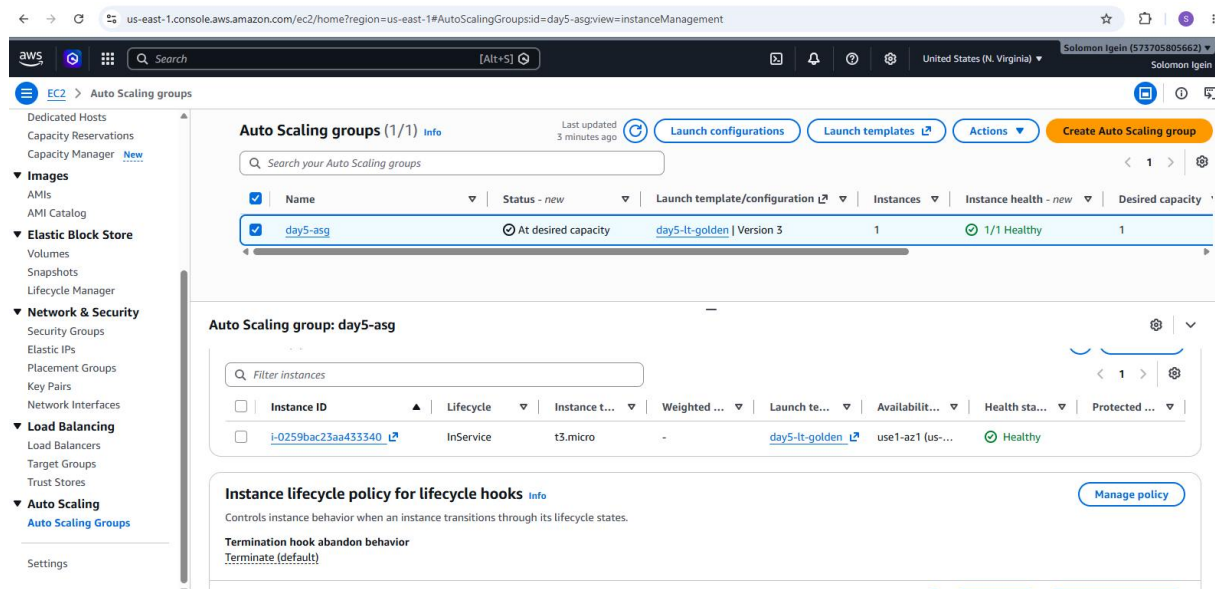


Fig 2: ASG day5-asg At desired capacity Version 3 1/1 Healthy — i-0259bac InService Healthy — After Chaos

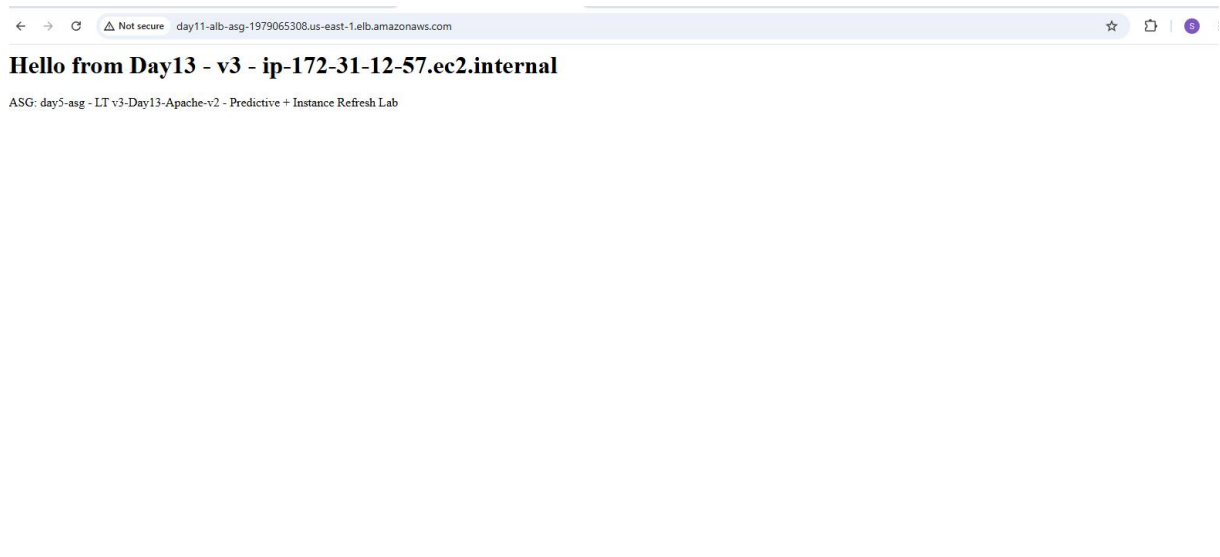


Fig 3: ALB day11-alb-asg-1979065308... → Hello from Day13 - v3 - ip-172-31-12-57.ec2.internal — Zero Downtime After Termination (NEW IP from Warm Pool)

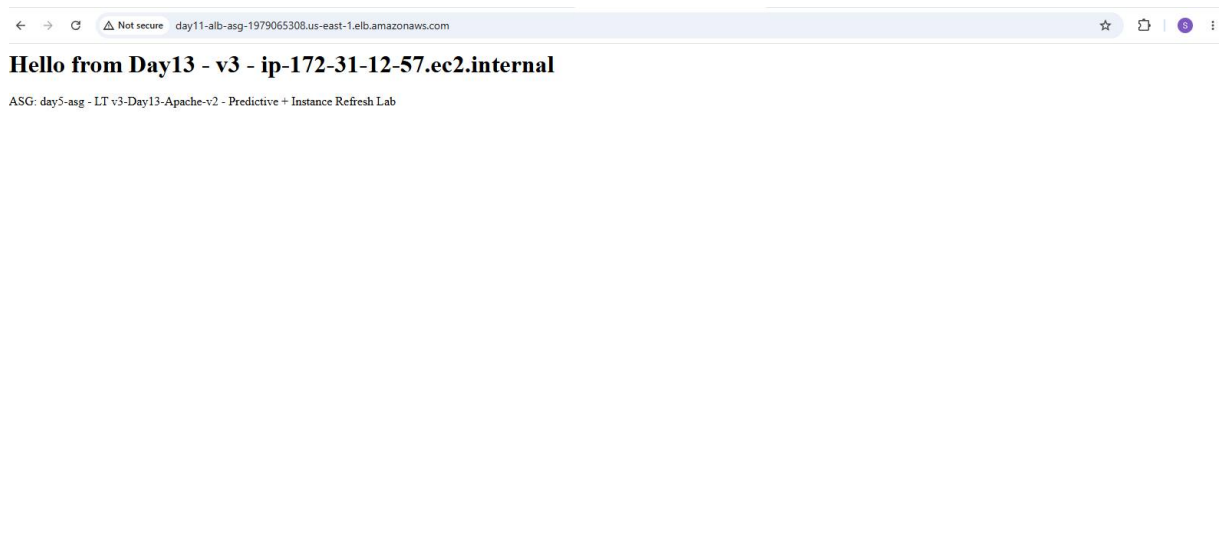


Fig 4: Same ALB V3 page second capture — Proves consistency

Interview Ready — 30 Sec Elevator Pitch + Cost & Cleanup

Elevator Pitch:

"I built enterprise Auto Scaling Group day5-asg with Launch Template Golden AMI Versioning V3, Instance Refresh rolling deployment a1e87d28 90% min healthy 300s warmup for zero downtime, Warm Pool 4 Stopped for 30s scale-out vs 90s cold (saving 90% cost vs Running), Chaos Engineering terminating i-0b11b05020b8d2046 and proving auto-heal from warm pool i-0259bac23aa433340 with ALB day11-alb-asg-1979065308 still serving Day13 V3 page with no downtime, Predictive scaling cpu-predictive-50 ML forecast + Scheduled 8am/8pm, Target Group Healthy checks, Lifecycle hooks, Secure ALB-only SG pattern."

Cost: 1 Running t3.micro \$7.60 + 4 Stopped EBS 8GB \$2.56 = \$10.16/mo portfolio cost. Optimized to 1 warm = \$8.40/mo.

Cleanup Order: 1) Delete Scheduled actions 2) Delete Warm Pool (Actions → Delete) 3) Delete ASG (check delete instances) 4) Delete ALB + TG 5) Terminate day2-test-server 6) Delete LT.

Next Steps: Add WAF to ALB, HTTPS ACM cert, CI/CD via CodeDeploy with Instance Refresh, Mixed Instances + Spot 70%, Capacity Rebalancing.

Verification Links: ALB DNS: <http://day11-alb-asg-1979065308.us-east-1.elb.amazonaws.com> | Instances: i-0259bac23aa433340 Running 34.231.241.165 | ASG: At desired capacity Version 3 1/1 Healthy

Signed: Solomon Igein — Day 5-14 Enterprise ASG Architect — 2026-08-08 — Status: 100% COMPLETE ■■■